

Linear Regression

HW 1 – Solutions

Directions: Submit a .pdf file containing your responses for the homework to Gradescope. The .pdf can be converted from Latex file, pictures of your handwriting solutions, word files, markdown files, etc (anything that can be converted into .pdf).

1. Consider the following statement: “To obtain the closed form expressions of the least squares estimators, it is required that the distribution of Y be normal.” Is this statement true or false, and why?

Sol: False. The solutions for the LS estimates only requires that $E(\epsilon_i) = 0$, $Var(\epsilon_i) = \sigma^2$, and $Cov(\epsilon_i, \epsilon_j) = 0$ for $i \neq j$, regardless of the precise distribution of ϵ_i (and therefore y_i). The Normality assumption is an additional assumption added in order to derive exact inferential results.

2. When there is a Normal distribution assumption on the error terms, we can also formulate Maximum Likelihood Estimators for β_0 , β_1 , & σ^2 . Use the likelihood function below to find the estimators $\hat{\beta}_0$, $\hat{\beta}_1$, & $\hat{\sigma}^2$, and show that the estimators β_0 , β_1 , are the same as the Least Square estimators. You do not need to check second derivatives to prove the maximum values. Likelihood function:

$$\mathcal{L}(\beta_0, \beta_1, \sigma^2) = \frac{1}{(2\pi\sigma^2)^{n/2}} \exp \left[-\frac{1}{2\sigma^2} \sum_{i=1}^n (Y_i - \beta_0 - \beta_1 X_i)^2 \right]$$

Sol: Under the assumption that $\epsilon_i \stackrel{iid}{\sim} \mathcal{N}(0, \sigma^2)$, $i = 1, \dots, n$ it follows from the OLS model that $y_i \stackrel{iid}{\sim} \mathcal{N}(\beta_0 + \beta_1 x_i, \sigma^2)$, $i = 1, \dots, n$. The likelihood for this sample is then:

$$\begin{aligned} L(\beta_0, \beta_1, \sigma^2) &= \prod_{i=1}^n (2\pi\sigma^2)^{-1/2} \exp \left\{ \frac{-1}{2\sigma^2} (y_i - \beta_0 - \beta_1 x_i)^2 \right\} \\ &= (2\pi\sigma^2)^{-n/2} \exp \left\{ \frac{-1}{2\sigma^2} \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)^2 \right\}, \end{aligned}$$

and so the loglikelihood is:

$$\ell(\beta_0, \beta_1, \sigma^2) = \frac{-n}{2} \log(2\pi\sigma^2) - \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)^2.$$

Notice that maximizing $\ell(\beta_0, \beta_1, \sigma^2)$ is totally equivalent to minimizing the sum of squares criterion $\sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)^2$ - therefore the maximum likelihood solutions for the intercept and slope must be the same as the LS solutions.

Alternatively, taking derivatives with respect to each parameter and setting equal to zero yields:

$$\begin{aligned} \text{i. } \frac{\partial \ell}{\partial \beta_0} &= \frac{-1}{\sigma^2} \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i) = 0, \\ \text{ii. } \frac{\partial \ell}{\partial \beta_1} &= \frac{-1}{\sigma^2} \sum_{i=1}^n x_i (y_i - \beta_0 - \beta_1 x_i) = 0, \end{aligned}$$

$$\text{iii. } \frac{\partial \ell}{\partial \sigma^2} = \frac{-n}{2\sigma^2} + \frac{1}{\sigma^4} \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)^2 = 0.$$

Equations i. and ii. are equivalent to the Normal equations which arise from solving the least squares problem, and thus the MLEs of slope and intercept must be the LS solutions.

For the variance estimate, solving iii. and plugging in the MLEs for slope and intercept yields:

$$\hat{\sigma}_{(MLE)}^2 = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2.$$

3. The solution for the LS estimator of the slope in simple linear regression is:

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n x_i y_i - n \bar{x} \bar{y}}{\sum_{i=1}^n x_i^2 - n \bar{x}^2}.$$

Prove that this expression is equivalent to the alternate formulation:

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2} = \frac{SSXY}{SSX}.$$

Sol: Starting with the solution's numerator, we see:

$$\begin{aligned} SSXY &= \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}) = \sum_{i=1}^n (x_i y_i - x_i \bar{y} - \bar{x} y_i + \bar{x} \bar{y}) \\ &= \sum_{i=1}^n x_i y_i - n \bar{x} \bar{y} - n \bar{x} \bar{y} + n \bar{x} \bar{y} \\ &= \sum_{i=1}^n x_i y_i - n \bar{x} \bar{y}. \end{aligned}$$

Then moving to the denominators:

$$\begin{aligned} SSX &= \sum_{i=1}^n (x_i - \bar{x})^2 = \sum_{i=1}^n (x_i^2 - 2\bar{x}x_i + \bar{x}^2) \\ &= \sum_{i=1}^n x_i^2 - 2n\bar{x}^2 + n\bar{x}^2 \\ &= \sum_{i=1}^n x_i^2 - n\bar{x}^2. \end{aligned}$$

Since the numerators and denominators are both equal, we have shown that

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n x_i y_i - n \bar{x} \bar{y}}{\sum_{i=1}^n x_i^2 - n \bar{x}^2} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2} = \frac{SSXY}{SSX}.$$

4. Recall that the residual for the i^{th} observation is defined as $e_i = y_i - \hat{y}_i$. Prove that $E(e_i) = 0$.

Sol:

Observe that:

$$\begin{aligned}
E(e_i) &= E(y_i - \hat{y}_i) = E(y_i) - E(\hat{y}_i) \\
&= \beta_0 + \beta_1 x_i - E(\hat{\beta}_0 + \hat{\beta}_1 x_i) \\
&= \beta_0 + \beta_1 x_i - E(\hat{\beta}_0) - E(\hat{\beta}_1) x_i \\
&= \beta_0 + \beta_1 x_i - (\beta_0 + \beta_1 x_i) = 0,
\end{aligned}$$

since the LS estimators are unbiased.

5. (a) When asked to state the simple linear regression model, a student wrote:

$$E(Y_i) = \beta_0 + \beta_1 x_i + \epsilon_i.$$

Do you agree? Why or why not?

Sol: No. The simple linear regression model is $Y_i = \beta_0 + \beta_1 X_i + \epsilon_i$. When we take the expectation of Y_i , it becomes $E(Y_i) = \beta_0 + \beta_1 X_i$ (i.e. No more randomness).

- (b) Consider the classical simple linear regression model. Suppose that the true parameter values are $\beta_0 = 2$, $\beta_1 = 4$, and $\sigma^2 = 9$. State the distributions of Y at $x = 1, 2$, and 4 , and explain how you found them.

Sol: We know that $(Y|X = x) \sim \mathcal{N}(_, _)$ with the following parameters:

$$E(Y) = E(\beta_0 + \beta_1 x + \epsilon) = \beta_0 + \beta_1 x$$

$$Var(Y) = Var(\beta_0 + \beta_1 x + \epsilon) = \sigma^2.$$

So:

X	Y distribution
1	$Y \sim N(6, 9)$
2	$Y \sim N(10, 9)$
4	$Y \sim N(18, 9)$

6. Consider the Rotten Tomatoes movie rating example, with the fitted SLR line $\hat{y}_i = 34.5089 + 0.4461x_i$.

- (a) Interpret the slope and the intercept in the real-life context of the problem.

Sol: Slope: For every point increase in a critic's rating, we expect the audience's rating to increase by 0.4461 points on average.

Int: For a movie with a critic rating of zero, we expect the audience's rating to be approximately 34.5089 on average.

- (b) Suppose the *Borderlands* movie is about to be released and critics have given it a score of 8 (out of 100) on Rotten Tomatoes. Using the fitted simple linear regression line, what do we predict the audience rating will be?

Sol: We predict an audience rating of $\hat{y}_i = 34.5089 + 0.4461(8) = 38.0777$.

- (c) What does the SLR line prediction in (b) suggest about who the regression line thinks is the harsher judge: audiences or critics?

Sol: In this case, we the SLR line predicts that the critics are harsher, since they rated *Borderlands* lower than the expected audience rating.

- (d) Suppose *The Quiet Place: Day One* is about to be released and critics have given it a score of 86 (out of 100) on Rotten Tomatoes. Using the fitted simple linear regression line, what do we predict the audience rating will be?

Sol: We predict an audience rating of $\hat{y}_i = 34.5089 + 0.4461(86) = 72.8735$.

- (e) What does the SLR line prediction (d) suggest about who the regression line thinks is the harsher judge: audiences or critics?

Sol: In this case, we the SLR line predicts that the audience is harsher, since they rated The Quiet Place lower than the expected audience rating.

- (f) Consider your findings in (c) and (e). Provide a reasonable explanation as to how one can reconcile these two results.

Sol: Since we have ground truth data for critic ratings, we frame the answer in terms of critically-unfavorable or critically acclaimed movies. Specifically, the SLR line expects that for critically-unfavorable movies (low critic ratings) the audience will enjoy them more and rate them higher. On the other hand, the SLR line expects that for critically-acclaimed movies (high critics ratings), audiences generally enjoy them less and on average do not give as high ratings as critics. Whether the audience is a harsher judge than critics ultimately depends on the type of movie it is in terms of critical acclaim. The phenomenon of the predicted response “regressing” back towards the mean audience rating instead of staying as extreme as the predictor is relative to its mean is the origin of the name “regression.”

- (g) What value of critic ratings will the SLR model predict the exact same score for audience ratings? Derive a general formula for this value in terms of $\hat{\beta}_0$ & $\hat{\beta}_1$.

Sol: We are searching for x such that $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x = x$.

So,

$$\begin{aligned}\hat{\beta}_0 + \hat{\beta}_1 x &= x \\ \Rightarrow \hat{\beta}_0 &= x(1 - \hat{\beta}_1) \\ \Rightarrow x &= \hat{\beta}_0 / (1 - \hat{\beta}_1),\end{aligned}$$

which in this case corresponds to a critic rating of approximately

$$x = 34.5089 / (1 - 0.4461) = 62.3017 \text{ points.}$$